Kunaal Agarwal

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Education

University of Virginia — M.D. Doctor of Medicine

Charlottesville, VA | 2025-2029

University of Virginia — B.A. Biology, B.A. Computer Science

Charlottesville, VA | 2021-2025

MCAT: 526, ACT: 35

Honors: Burhan Chaudhry Scholarship, Harrison Research Award, Jefferson Trust Grant.

Formative Coursework: Data Structures and Algorithms, Discrete Math, Foundations of Data Analysis, Cell Biology, Organic Chemistry, Molecular Biology & Genetics, Biochemistry, Evolution & Ecology.

Self-Study: Linear Algebra, Probability Theory, Game Theory, Machine Learning, Deep Learning.

Works

1. **Presentation: Agarwal, K.**, Henry, T. R., Van Horn, J. D., & Rasero, J. (2026). **"niBuild: Towards FAIR multi-modal neuroimaging analysis"** *UVA Brain Symposium.* Poster
2. **Presentation: Agarwal, K.**, Clark, M., & Doryab, A. (2025). **"Moodring: A Machine Learning Framework for Depression Prediction"** *UVA Undergraduate Research Symposium.* Poster
3. **Agarwal, K.**, Kim, H. R., Almeida, J. S., & Sandoval, L. (2023). **"MedicaidJS: A FAIR approach to real-time drug analytics."** *Bioinformatics Advances*, **3**(1), vbad170. DOI
   (a) **Presentation: Agarwal, K.** & Kim, H. R. (2023). **"Medicaid.js, a FAIR Approach to Real Time Drug Analytics."** *NIH Summer Internship Program Poster Day.* Poster

Research

Student Researcher

University of Virginia | Charlottesville, VA | Jan 2024 - Present

- Advisor: **Javier Rasero, PhD**, School of Data Science.

Project: Infrastructure for multimodal neuroimaging analysis.

- **Motivation**: Neuroimaging is becoming increasingly accessible, but despite progress in open-source paradigms for dissemination, analysis remains fragmented.
- **Research**: We developed a graphical user interface (GUI) enabling users to visually construct multimodal neuroimaging workflows and produce reproducible, shareable workflows implemented in the Common Workflow Language (CWL). The goal of the software is to create a platform emphasizing the FAIR standards for data stewardship and provenance. We also explored popular paradigms for isolating execution of workflows including containerization on high performance computing environments (HPC).

Student Researcher

University of Virginia | Charlottesville, VA | Aug 2023 - Jan 2026

- Advisor: **Afsaneh Doryab, PhD**, Department of Computer Science.
- The **Human-AI Lab** studies human behavior through computational modeling.

Project — Online learning models to predict future depressive episodes and symptom severity changes in clinically depressed adolescents using passively sensed data streams from mobile device and wearable technology sensors.

- **Study Design**: ∼50 clinically depressed adolescents were enrolled in the second iteration of a 6-month longitudinal study where we passively collected data from their smartphone and provided Fitbit. We additionally collected weekly patient health questionnaires (PHQ-8) which is the gold-standard clinical algorithm for depression diagnosis.
- **Data preparation**: Data streams were derived from sensors inherent to mobile devices, including GPS, accelerometer, battery consumption, screen time, etc. We installed the AWARE framework onto participants' smartphones which enabled efficient logging of the raw sensor data into our database. We then transformed the raw data into meaningful feature representations via the RAPIDS framework.
- **Research**: Concerned by invasiveness, we explored modeling techniques to optimize data privacy. We analyzed model capabilities in predicting dependent variable constructs of varying difficulties and and then performed a series of ablation studies. The goal was to learn the bare minimum data representation required to make useful predictions.

Summer Research Fellow

National Cancer Institute (NCI, NIH) | Rockville, MD | May 2023 - Aug 2023

- Advisor: **Jonas Almeida, PhD**, Division of Cancer Epidemiology and Genetics.
- The **Data Science and Engineering Group** builds software infrastructure for healthcare data repositories.

Project: Infrastructure for interaction with drug metric data APIs.

- **Motivation**: Large healthcare data repositories exist but are underused by non-technical researchers due to their esoteric interfaces. We sought to enable medical professionals, amongst others, to leverage useful data repositories and make more informed clinical decisions.
- **Research**: We developed an end-to-end JavaScript SDK for data aggregation, analysis, and visualization of drug-metric data repositories, including Data.medicaid, RxNorm, and FDA APIs. Designed a lightweight client-side architecture for seamless integration and installation. Ensured FAIR compliance, optimizing data for findability, accessibility, interoperability, and reusability. We published the project as an NPM package and ObservableHQ notebooks for documentation and prebuilt tools.

Organizations

President & Founder — Clean Plate Project

University Of Virginia | Charlottesville, VA | Jan 2024 - May 2025

The **Clean Plate Project** is a humanitarian organization at the intersection of food and medicine.

- Our mission is to provide support to patients and their families at the University of Virginia Children's Hospital by serving authentic, home-cooked meals during their time of need.
- **Impact** — Established relationships with the Ronald McDonald House and the Yellow Door Foundation, organizations which provide long-term housing for families of patients at the Children's Hospital. Since founding, expanded the organization to 250 members and hosted dozens of cooking sessions.
- Featured within the **Cavalier Daily newspaper** and **Virginia Audio Collective podcast**

President & Founder — Virginia Poker Club & Cavalier Competitive Poker

University Of Virginia | Charlottesville, VA | Aug 2023 - May 2025

The **Virginia Poker Club (VPC)** is UVA's official poker organization and **Cavalier Competitive Poker (CCP)** is our competitive team.

- VPC fosters a community centered around poker theory, decision theory, and risk-taking. Our mission is to create a friendly, zero-pressure environment where students can learn and enjoy the beautiful game of poker.
- **Impact** — Organized weekly meetings, tournaments, and industry-sponsored events with trading firms such as Susquehanna Investment Group (SIG) and Flow Traders, engaging over 250 members. Delivered strategy presentations covering fundamental skills (ex. position, ranges) and advanced poker theory (ex. equity realization factor, indifference).
- **CCP** — We compete in the **Intercollegiate Poker Association (IPA)** against other universities on an international scale.
  - **2024**: Team captain, Regular season group champions, 3rd seed overall (out of 37 teams), 2x individual 1st place weekly finishes.
  - **2025**: Playoff qualifier via wildcard victory, 15th seed overall out of (36 teams).